



# **CENTER FOR INTEGRATED BUILDING SYSTEMS**

College of Engineering, Architecture and Technology

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## **CNN–based Prediction of Boiling Heat Transfer**

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# Introduction to Boiling

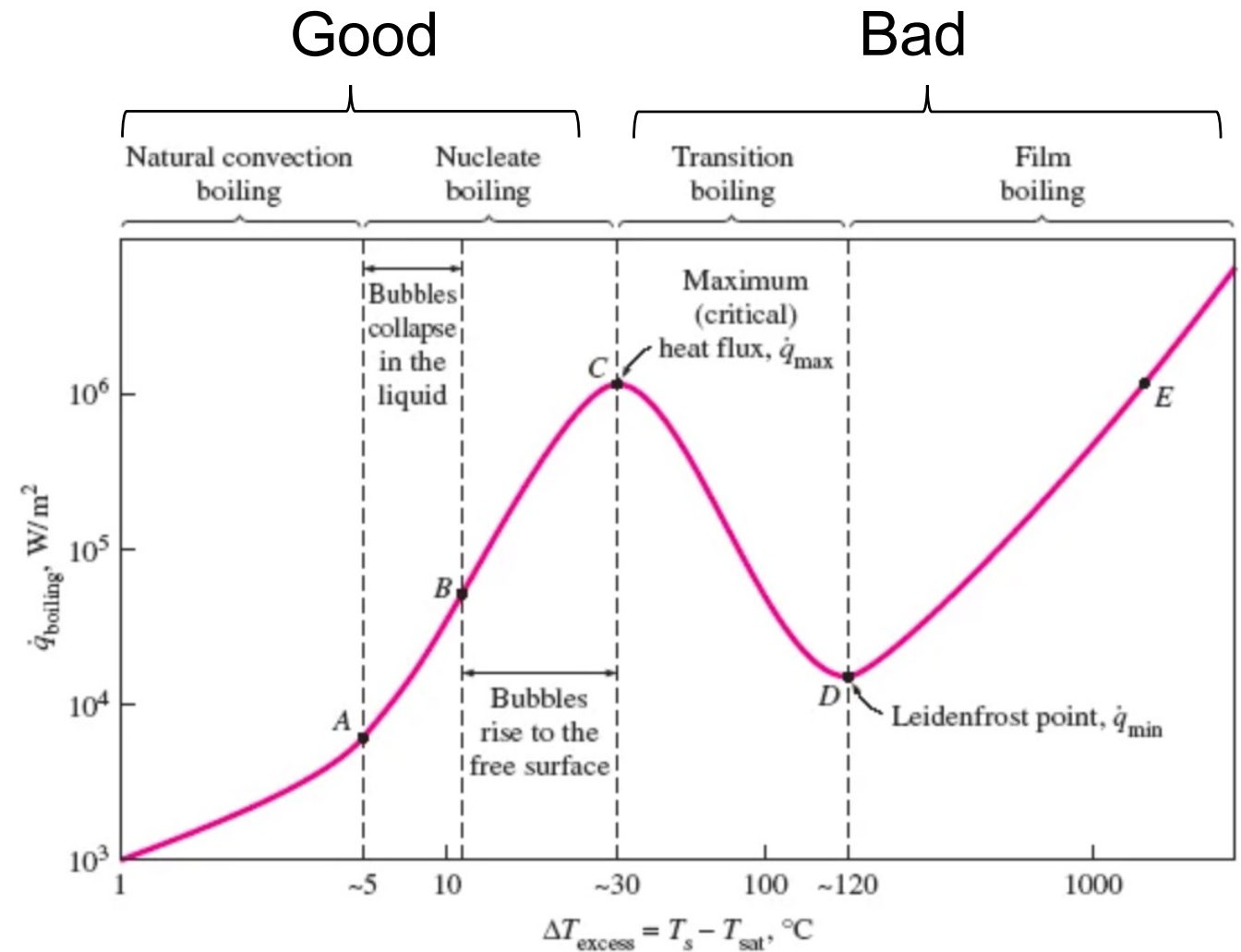
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- Heat transfer is a limiting design factor in many high-energy systems
- Typical cooling strategies include forcing air/liquid over a heat source
  - » Convective heat transfer can carry energy away from heat sources
- Much more effective to boil a cooling fluid and take advantage of latent heat
  - » 10-100x better heat transfer



# Introduction to Boiling

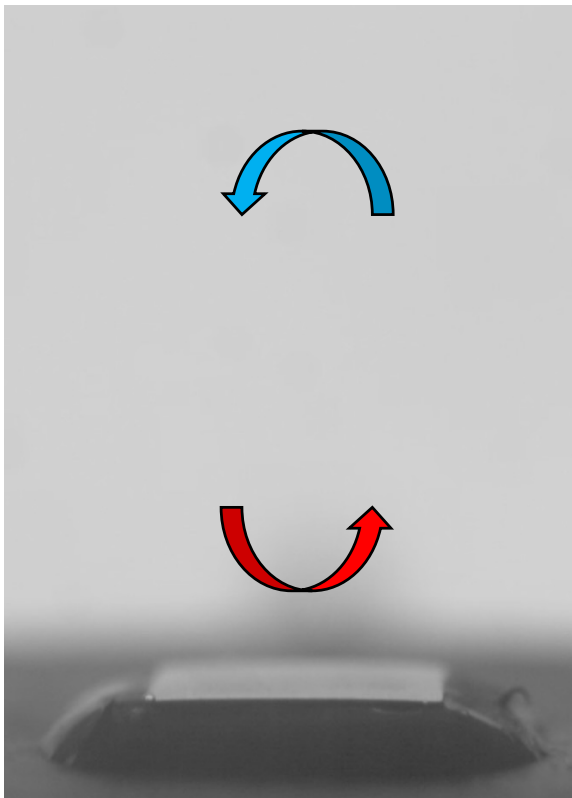
- Plot of boiling heat transfer
  - » Y-axis is heat transfer
  - » X-axis is temperature difference between fluid and surface
- Boiling takes place in different “regimes” based off conditions
  - » Early regimes are more stable and safer to operate in, but relatively low heat transfer
  - » Late regimes offer much higher heat transfer, but very unstable and unpredictable



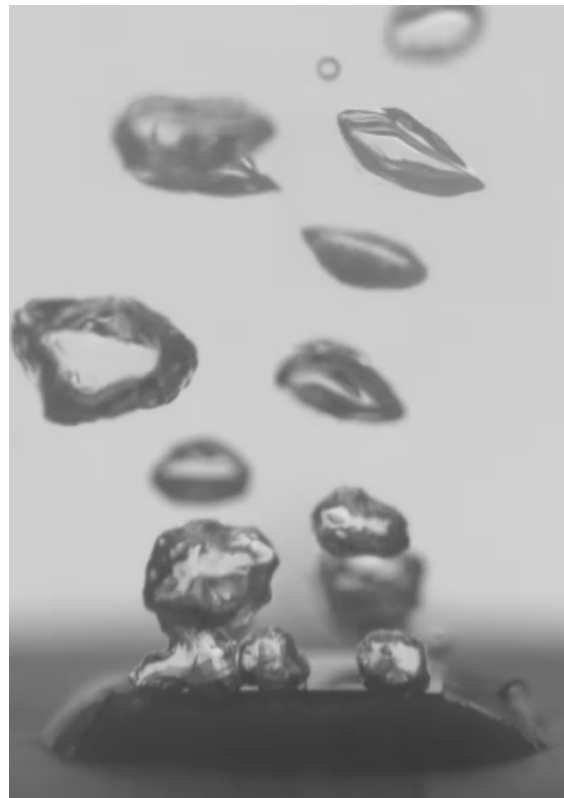
# Introduction to Boiling

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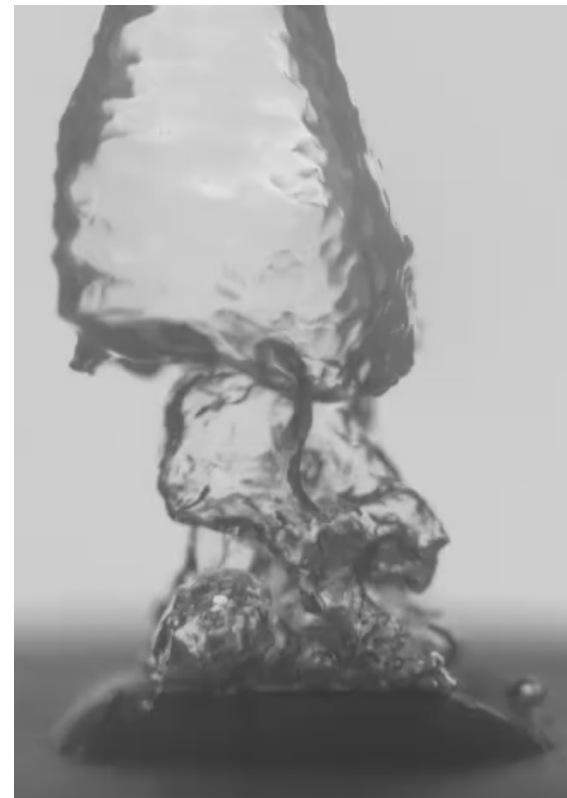
Natural  
Convection  
Boiling



Early Nucleate  
Boiling



Late Nucleate  
Boiling



Near CHF  
Boiling



# Introduction to Boiling

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- Video shows unstable bubble growth and departure as boiling approaches critical heat flux
- Visually different to early boiling regimes
- Visual differences open opportunities to use CNNs



# Target Metric for CNN

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- As mentioned, it is beneficial to operate near CHF but important to understand instabilities to operate safely
- Existing CNN-based work exists to predict heat transfer and CHF, but we define a new metric: **CHF Proximity**

$$\Pi_{CHF} = \frac{T_{wall} - T_{sat}}{(T_{wall} - T_{sat})_{CHF}}$$

# About the dataset

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- Data from: “Unveiling hysteresis of transient boiling: A multimodal perspective”
  - » Published by Pandey et. al, in 2025
  - » Study to understand how boiling is different when ramping up and down in power using visual, audio, and temperature data
- Produced and published two full videos of boiling ramping up to CHF and back down, along with all temperature data
  - » Boiling\_84: Deionized water on a copper foam surface
  - » Boiling\_91: Deionized water on a smooth copper surface

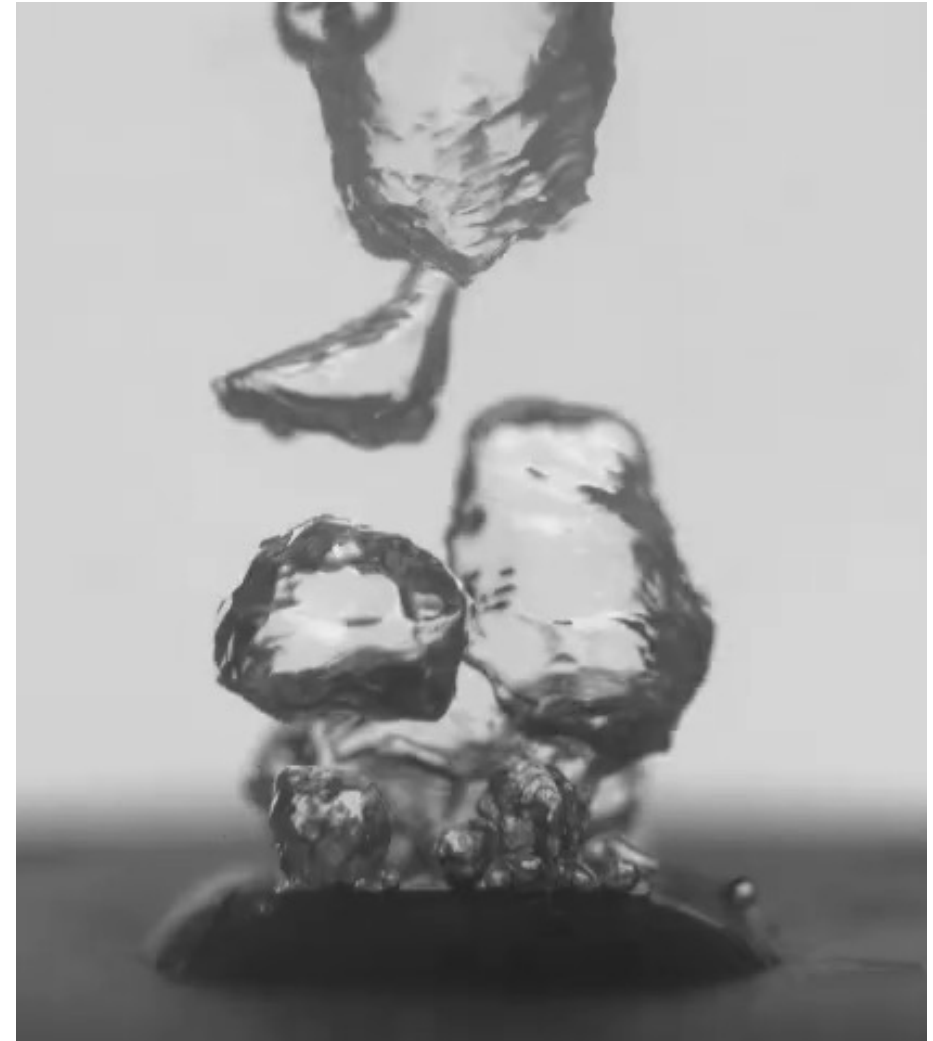
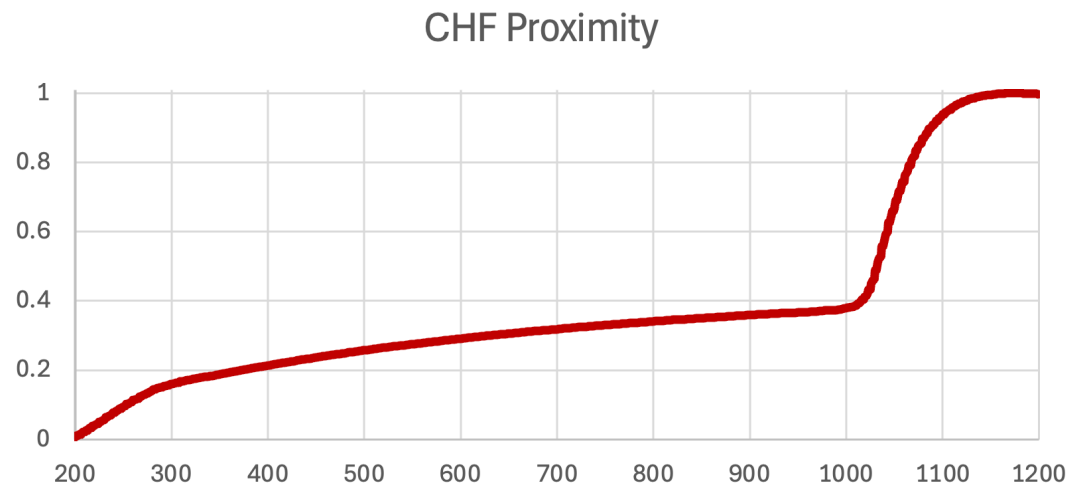
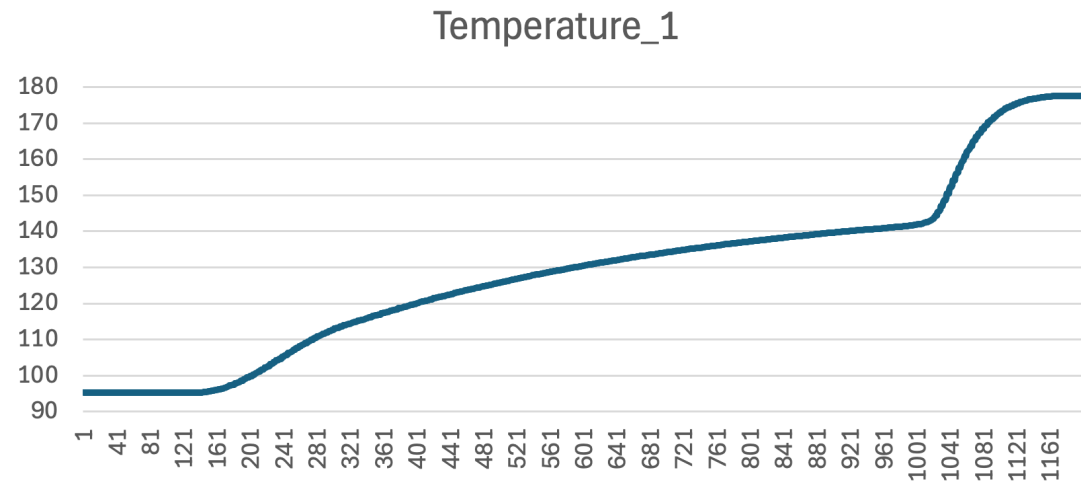
# Data Processing

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- Needed to create dataset of images and temperature points that are lined up
  - » Boiling\_84: 45 minutes at 60fps, 1.5 million temperature points
  - » Boiling\_91: 32 minutes at 60 fps, 1 million temperature points
- Original dataset was far too data dense, so down-sampling was done
- Python script created to chop up raw data and created linked image frame at every temperature point
- Resulting in 1200 data points from each set from zero boiling up to CHF



# Data Processing



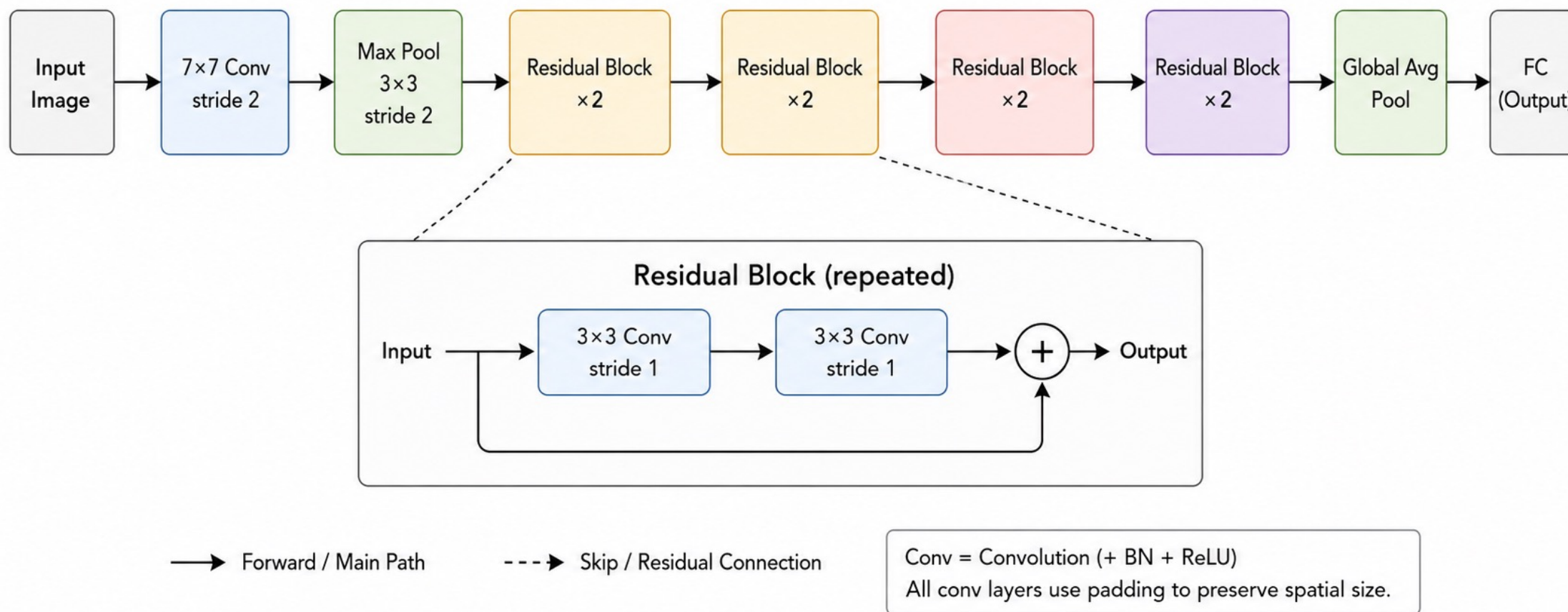
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- Resulting in 1200 data points from each set from zero boiling up to CHF
- One set used for training, other set used for testing

# CNN Network Choice

- Decided to go with ResNet-18 (reasoning shown in next slides)



# CNN Network Choice

- Dataset is relatively small, likely insufficient to train a complex model
- Chose to use a pretrained model, perform training on pretrained weights

- » AlexNet
- » ResNet
- » InceptionNet
- » Etc.

| CNN Architecture | Model F1 Score |
|------------------|----------------|
| Basic CNN        | 91.11%         |
| <b>AlexNet</b>   | <b>98.87%</b>  |
| <b>ResNet</b>    | <b>96.50%</b>  |
| InceptionNet     | 94.23%         |

- Classification of boiling regimes, fluids, and heating surfaces through deep learning algorithms and image analysis
  - Published by Paz et. al, in 2025

# Testing different networks

- Results of my own network experimentation:

| CNN Architecture | Best MAE (%) | RMSE (%)    | $R^2$        | Correlation  |
|------------------|--------------|-------------|--------------|--------------|
| AlexNet          | 13.5         | 23.6        | 0.429        | 0.789        |
| ResNet-18        | <b>11.3</b>  | 21.9        | 0.505        | 0.792        |
| ResNet-34        | 13.9         | 24.4        | 0.388        | 0.737        |
| EfficientNet     | 12.6         | <b>21.7</b> | <b>0.514</b> | <b>0.796</b> |

# Tuning ResNet-18 Model

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Different ways I tried to improve the model:

1. Training Hyperparameter Tuning:

- Batch size optimal at 32
- Learning rate best at  $1e-4$
- Weight Decay at  $1e-4$
- Backbone not frozen during training

14.7% MAE



2. Data Augmentation

- Color Jitter made very small improvement
- Random scaling and translation made

11.3% MAE

3. Output layer dropout

- Addition of dropout = 0.2 beneficial, 0.4 too much

# Final Model and Results

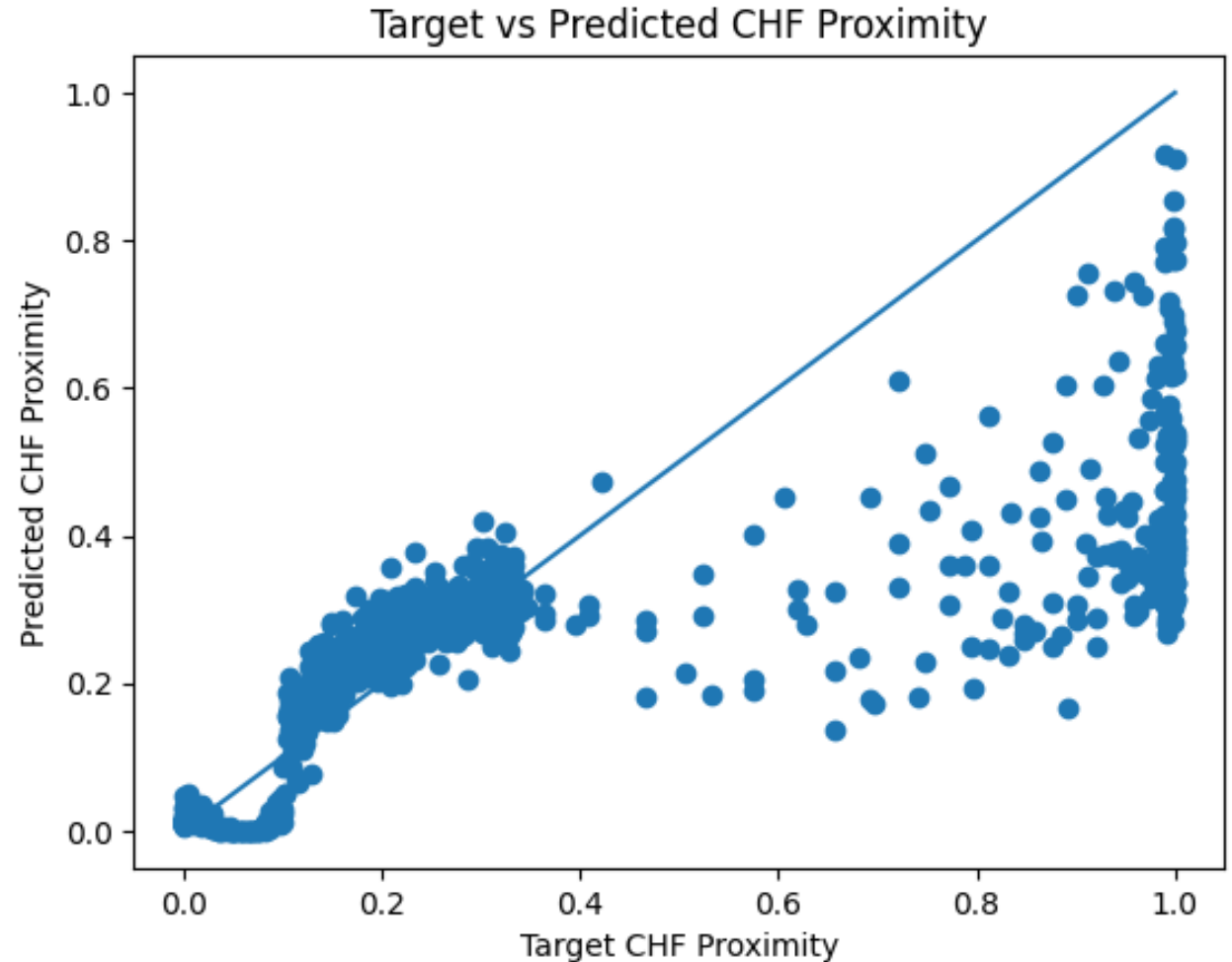
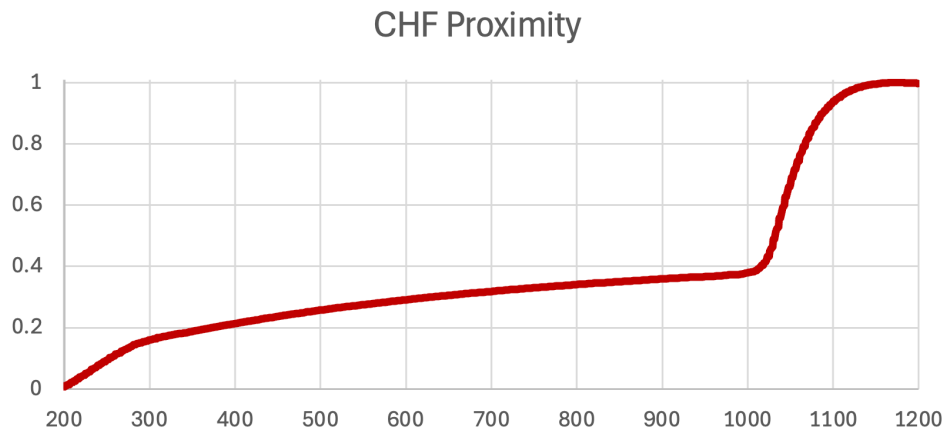
Final Model:

- ResNet-18 with Sequential Output Layer
  - One feedforward layer
  - Dropout: 0.2
- Epochs: 10
- Learning Rate: 1e-4
- Weight Decay: 1e-4
- Validation Split: 0.2

| Performance Metric |       |
|--------------------|-------|
| MAE (%)            | 11.3  |
| RMSE (%)           | 21.9  |
| $R^2$              | 0.505 |
| Correlation        | 0.792 |

# Final Model and Results

- Target vs Predicted for ResNet-18
- Shows very strong prediction for targets between 0.1 to 0.4
- Very poor performance above 0.5 due to lack of data





# Conclusions and Future Work

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- ResNet-18 outperformed other tested networks
  - Hyperparameters tuned to optimize performance
  - Data augmentation improved generalization
  - Dropout improves generalization
- Shortage of training data
  - Model was bound to struggle with 50/50 train/test split
- Need to account for data imbalance
- Investigate dynamic CNN models
  - Leverage time series stream of images

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# Thank you for listening!

